

A chaotic double pendulum in MuJoCo

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The double pendulum is the textbook example of a deterministic system that is, in practice, unpredictable: two arms hinged in series have four state variables $(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$ and a Lagrangian with one nonlinear coupling term, and that is enough for trajectories that start arbitrarily close together to separate exponentially. We ran two identical pendulums side by side, started 10^{-3} rad apart, and measured when and how far they diverge.

Methods

The demo runs two double pendulums at once, side by side in the same MuJoCo scene, with identical mass, length, and damping. The only difference is the initial angle of the upper arm,

$$\theta_1^A(0) = 2.0, \quad \theta_1^B(0) = 2.0 + 10^{-3},$$

in radians. The MJCF model is two double-pendulum chains:

- two chains of two hinge arms each, anchored at $x = \pm 0.6$ m
- arms are 0.4 m capsules; a small site marks each tip so its world position can be read
- timestep 0.002 s with the RK4 integrator, important in the chaotic regime, where forward Euler would let small numerical errors swamp the actual divergence
- damping 0, so total energy is conserved (give or take RK4’s drift)

Tip separation is measured in each pendulum’s *anchor-local* frame (each tip position minus its own anchor) so the constant 1.2 m horizontal offset between the two scenes doesn’t contaminate the metric.

Results

For the first two seconds or so the two pendulums trace what looks like the same motion. Around $t \approx 3$ s the tip-to-tip distance crosses the 0.1 m threshold, and from then on they are visibly doing completely different things. The Lyapunov-like blow-up is the whole story.

[Video — Two near-identical double pendulums (started 10^{-3} rad apart) in one MuJoCo scene: they track each other for the first couple of seconds, then diverge visibly once the tip separation passes the 0.1 m threshold. · view the web edition to play.]

The maximum separation hits 1.21 m, basically the full extent of a single pendulum (0.8 m), meaning that at some point one tip is at the top of its swing while the other’s is at the bottom. The final separation of 0.56 m is just where they happen to be at $t = 8$ s; with a chaotic system, “where they end up” is meaningless past the divergence time.

Run parameters

Parameter	Value
theta1	2
theta2	0
epsilon	0.001
duration	8
fps	60
separation_threshold	0.1
separation_time_s	2.968
max_separation_m	1.2096309484781214
final_separation_m	0.5648867078342311

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